

5. PROJECT DEVELOPMENT PHASE

Medical Diagnosis Center by Service Portal (ServiceNow)

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The objective of the development phase is to implement the required Medical Diagnosis Center activities in ServiceNow and demonstrate the complete application flow. The development includes application creation, patient registration, diagnostic test management, appointment booking, appointment cancellation, refund calculation, report generation, and Service Portal access.

Development Flow

ServiceNow Application → Patient Registration → Diagnostic Test Management → Test Selection → Appointment Booking → Appointment Management → Appointment Cancellation → Refund Calculation → Appointment Completion → Automatic Report Generation → Report Management → Service Portal Access

5.1 Phase 1 — Application Creation

Create the Medical Diagnosis Center application in ServiceNow.

Implementation: Create the application in ServiceNow and configure the required application components for the Medical Diagnosis Center. The application provides the base environment for managing patients, diagnostic tests, appointments, and reports.

Result: The Medical Diagnosis Center application is created and ready for development.

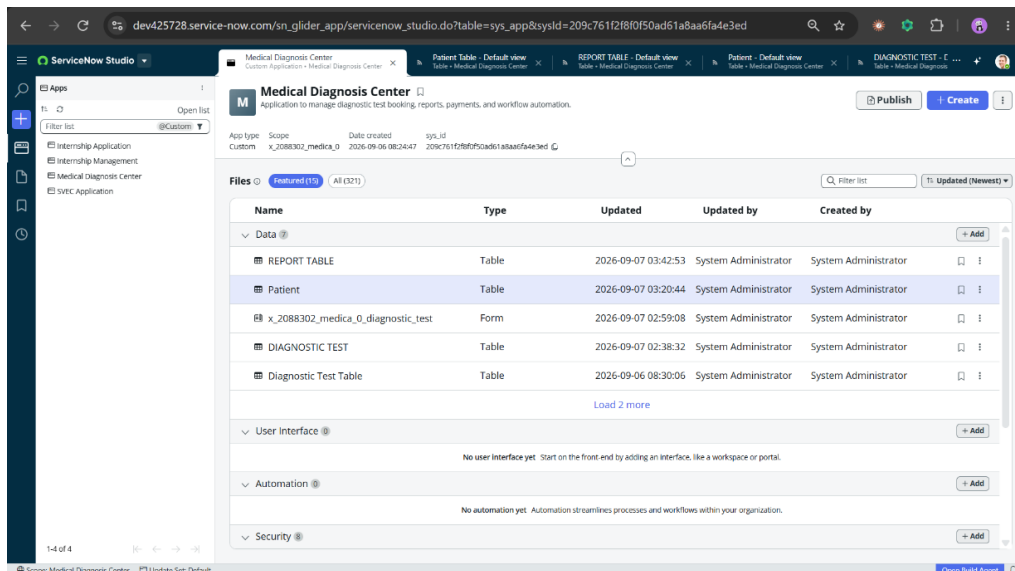


Figure 6: Medical Diagnosis Center ServiceNow Application

5.2 Phase 2 — Patient Table Development

Create the Patient table to store and manage patient information.

Implementation: Create the Patient table and configure the required fields for patient registration and management. The table is used to maintain patient records that can be associated with appointments.

Result: The Patient table is created and patient information can be maintained in the application.

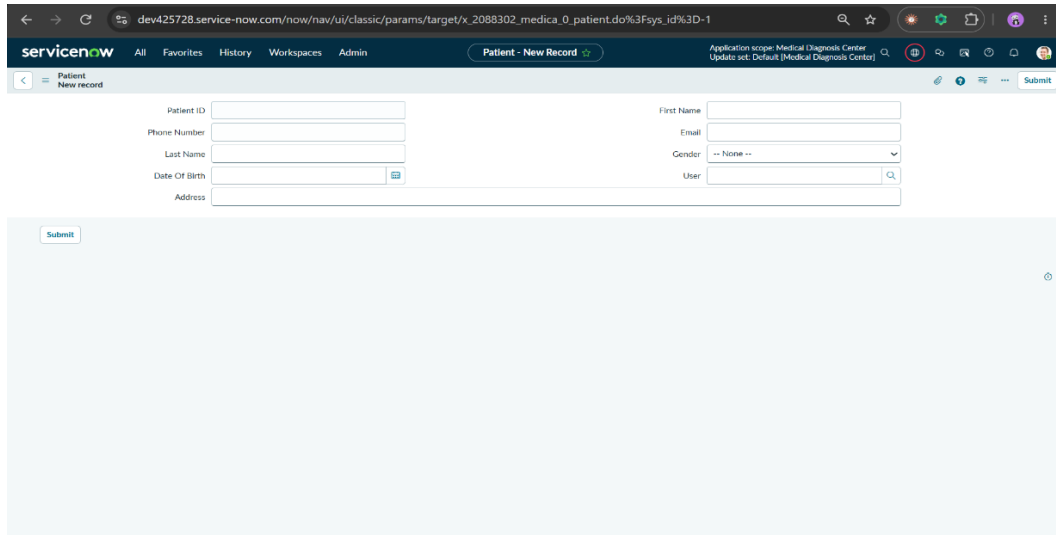
A screenshot of the ServiceNow 'Patient - New Record' form. The form is displayed in a web browser window with the URL 'dev425728.service-now.com/now/nav/ui/classic/params/target/x_2088302_medica_0_patient.do%3Fsys_id%3D-1'. The form includes fields for Patient ID, Phone Number, Last Name, Date Of Birth, Address, First Name, Email, Gender (a dropdown menu currently showing '-- None --'), and User (a dropdown menu with a search icon). A 'Submit' button is located at the bottom left of the form.

Figure 7: Patient Table

5.3 Phase 3 — Diagnostic Test Table Development

Create the Diagnostic Test table to maintain the diagnostic tests provided by the center.

Implementation: Create the Diagnostic Test table and configure the required fields for maintaining diagnostic test information. The test records are used when patients select a diagnostic test for an appointment.

Result: The Diagnostic Test table is created and diagnostic test information can be maintained.

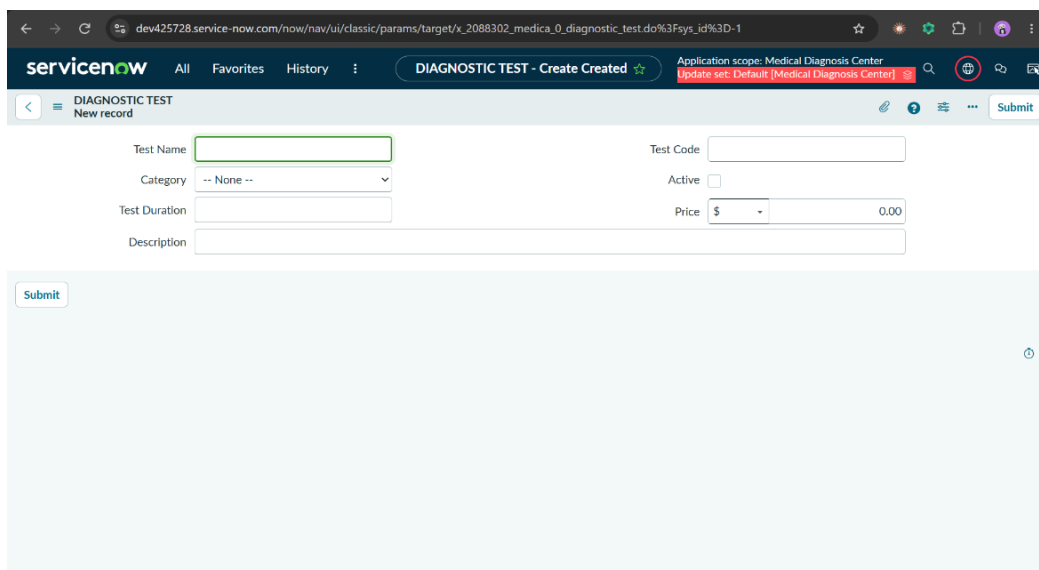
A screenshot of the ServiceNow 'DIAGNOSTIC TEST - Create Created' form. The form is displayed in a web browser window with the URL 'dev425728.service-now.com/now/nav/ui/classic/params/target/x_2088302_medica_0_diagnostic_test.do%3Fsys_id%3D-1'. The form includes fields for Test Name, Category (a dropdown menu currently showing '-- None --'), Test Duration, Description, Test Code, Active (a checkbox), and Price (a dropdown menu showing '\$' and a text field showing '0.00'). A 'Submit' button is located at the bottom left of the form.

Figure 8: Diagnostic Test Table

5.4 Phase 4 — Appointment Table Development

Create the Appointment table to manage patient appointments.

Implementation: Create the Appointment table and configure the required fields. Establish the required relationship between the patient and diagnostic test so that an appointment can contain the selected patient and diagnostic test information.

Result: The Appointment table is created and appointment records can be maintained.

The screenshot shows the ServiceNow interface for creating a new appointment record. The form is titled 'APPOINTMENT TABLE - Create New record' and includes a 'Submit' button. The form fields are organized into two columns. The left column contains: Appointment Number (text input), Patient (lookup field), Status (dropdown menu with '-- None --'), Approval Status (dropdown menu with '-- None --'), Email (text input), Approval Date (calendar icon), Test (lookup field), Scheduled Date (calendar icon), and Slot Time (dropdown menu with 'Configurable time slots'). The right column contains: Rejection Reason (text input), Confirmation Sent (checkbox), Created By (lookup field), Approved By (lookup field), Notes (text input), Payment Status (dropdown menu with '-- None --'), Total Amount (currency field with '\$' and '0.00'), and Refund Amount (currency field with '\$' and '0.00'). A 'Submit' button is located at the bottom left of the form.

Figure 9: Appointment Table

5.5 Phase 5 — Report Table Development

Create the Report table to maintain diagnostic reports related to completed appointments.

Implementation: Create the Report table and configure the required fields for report management. The report record is associated with the relevant appointment so that the generated report can be maintained against the appointment.

Result: The Report table is created and report information can be managed.

The screenshot shows the ServiceNow interface for creating a new report record. The form is titled 'REPORT TABLE - Create New record' and includes a 'Submit' button. The form fields are organized into two columns. The left column contains: Patient (lookup field), Report File (text input with 'Recent selections' and 'Phani Teja Sri'), and Findings (text input with a 'Findings' icon). The right column contains: Appointment (lookup field), Report Date (calendar icon), and Verified By (lookup field). A 'Submit' button is located at the bottom left of the form.

Figure 10: Report Table

5.6 Phase 6 — Business Rules and Automation

Configure the required business rules and automation for the Medical Diagnosis Center process.

Implementation: Configure the required automation for appointment management, appointment cancellation, refund calculation, and report generation after appointment completion. The configured rules perform the required actions based on the appointment process.

Result: The required business rules and automation are configured for the Medical Diagnosis Center application.

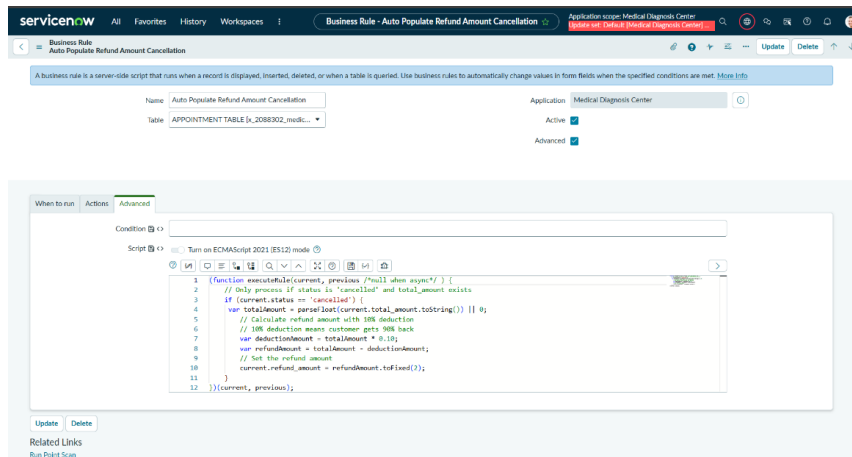


Figure 11.1: Business Rule — Auto Populate Refund Amount Cancellation

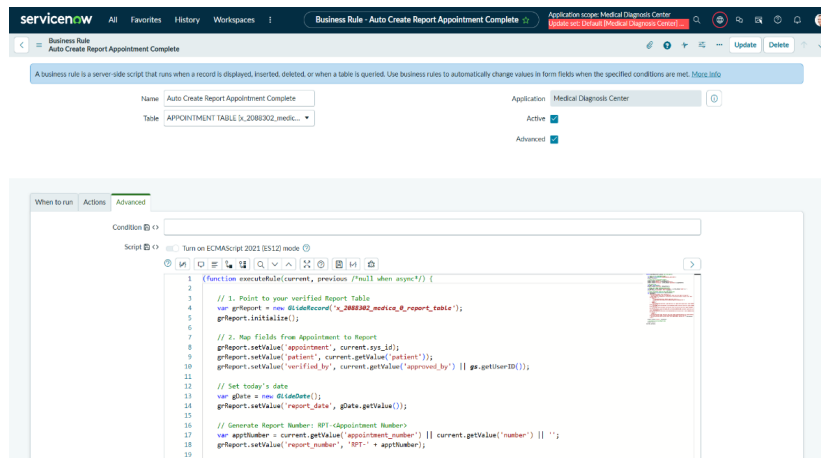


Figure 11.2: Business Rule — Auto Create Report Appointment Complete

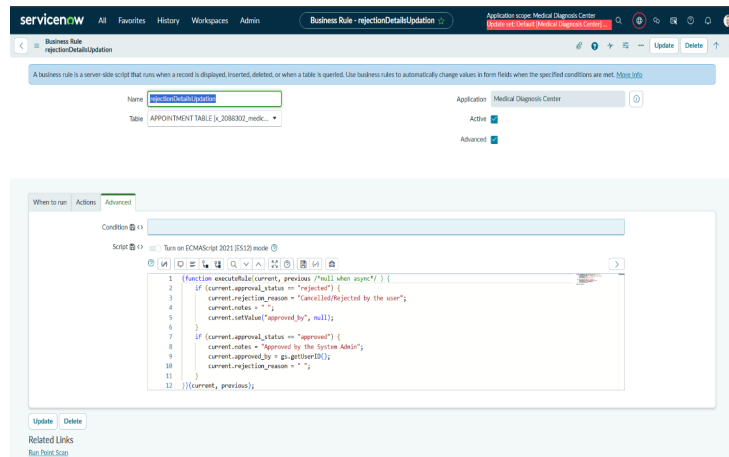


Figure 11.3: Business Rule — rejectionDetailsUpdate

5.7 Phase 7 — Service Portal Development

Develop the Service Portal to provide users with access to the Medical Diagnosis Center services.

Implementation: Create and configure the Service Portal and provide the required portal components for accessing the Medical Diagnosis Center functionality. The portal provides a simple interface for users to interact with the application.

Result: The Medical Diagnosis Center Service Portal is created and accessible to users.

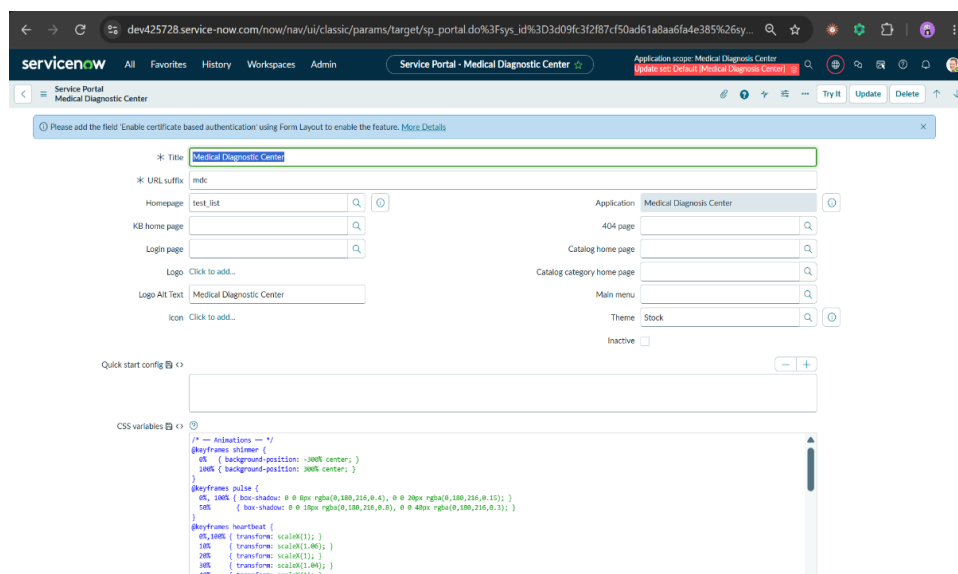


Figure 12: Medical Diagnosis Center Service Portal

5.8 Phase 8 — Portal Pages and Widgets

Configure the required pages and widgets for the Medical Diagnosis Center Service Portal.

Implementation: Create or configure the required portal pages and widgets for the application. The widgets provide the user interface required for accessing and performing the available Medical Diagnosis Center functions.

Result: The required Service Portal pages and widgets are configured and available to users.

5.10 Phase 10 — Appointment Cancellation and Refund

Implement appointment cancellation and refund calculation for cancelled appointments.

Implementation: Open the required appointment and perform the cancellation process. The configured system logic calculates the applicable refund for the cancelled appointment according to the project requirements.

Result: The appointment is cancelled and the required refund amount is calculated.

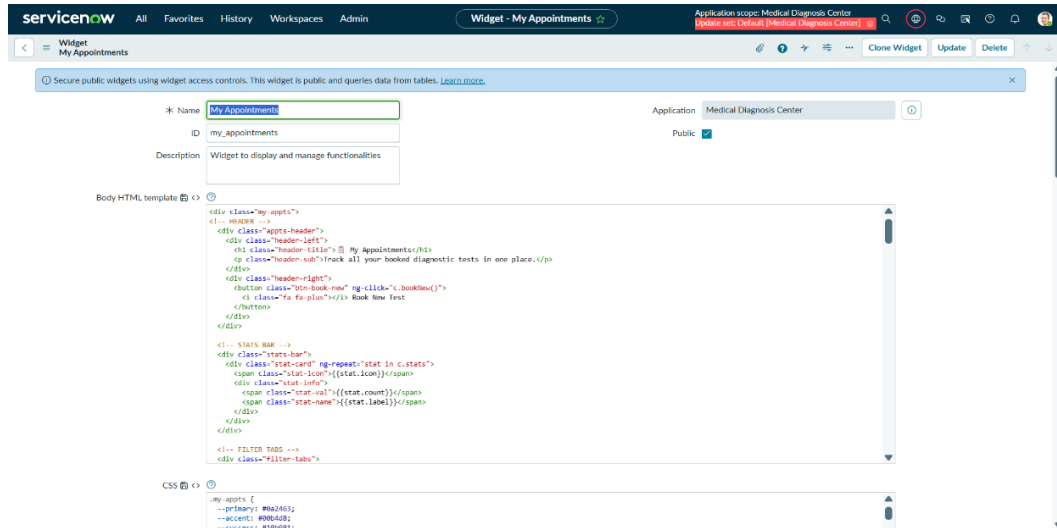


Figure 15: Appointment Cancellation and Refund Calculation

5.11 Phase 11 — Appointment Completion and Report Generation

Implement automatic report generation after completion of an appointment.

Implementation: Update the appointment to the completed state. The configured automation identifies the completed appointment and generates the corresponding report record.

Result: The diagnostic report is automatically generated for the completed appointment.

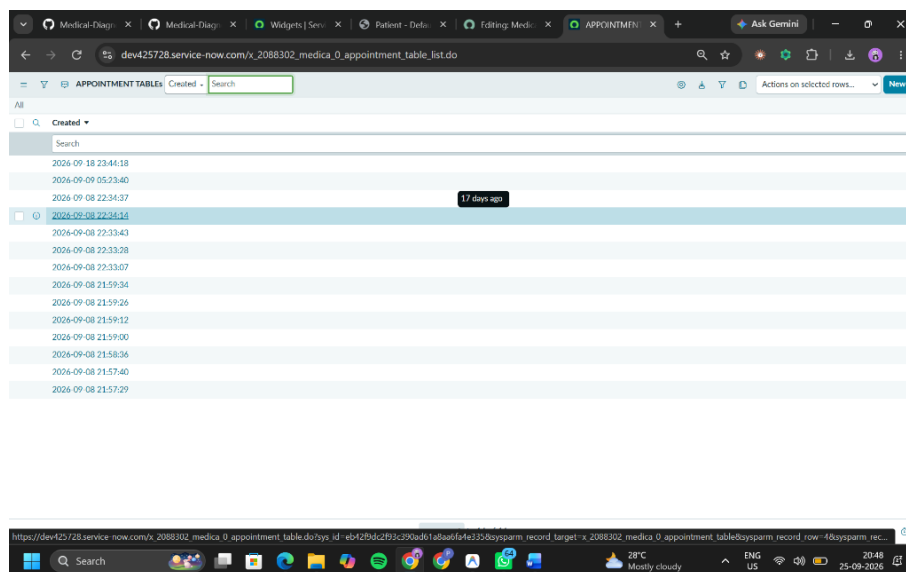


Figure 16: Automatic Report Generation — Completed Appointments Record Verification

5.12 Phase 12 — Report Management

Manage the reports generated for completed appointments.

Implementation: Open the generated report and verify the associated appointment and required report information. The report record can be maintained through the ServiceNow application.

Result: The generated report is available and associated with the relevant appointment.

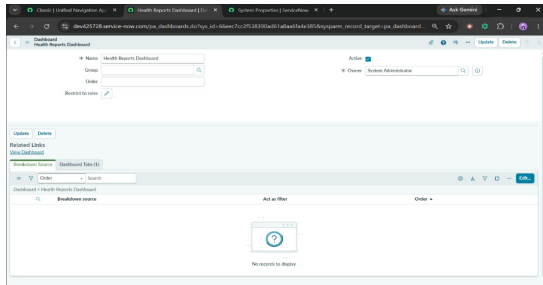


Figure 17.1: Report Management — Health Reports Dashboard

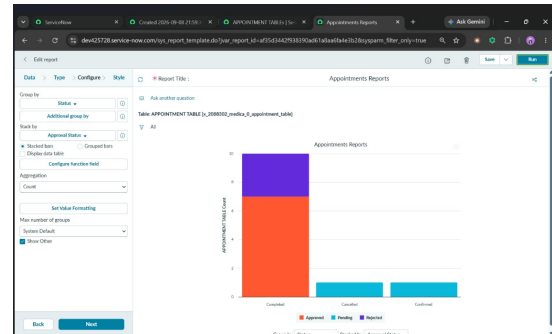


Figure 17.2: Report Management — Appointments Reports Analytics Chart

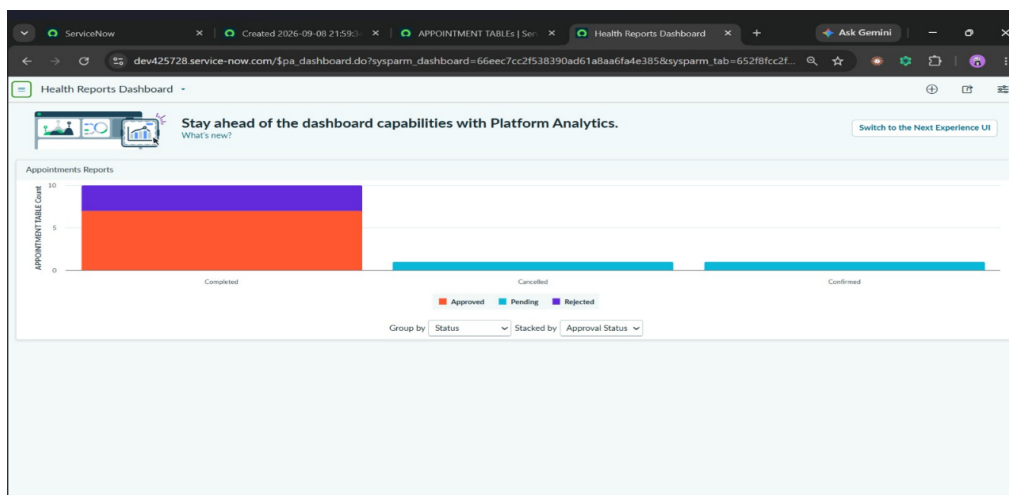


Figure 18: Testing & Deployment Analytics Dashboard

5.13 Phase 13 — Final Validation

The completed solution is checked to confirm that the main records and processes are working correctly.

- Patient registration is available.
- Diagnostic tests can be maintained and selected.
- Appointments can be created and managed.
- Appointment cancellation is supported.
- Refund calculation is performed for cancelled appointments.
- Completed appointments generate the required report.
- Reports can be managed.
- Service Portal access is available.

5.14 Phase 14 — Testing and Deployment Validation

The Medical Diagnosis Center application is tested end-to-end to confirm that the implemented activities work together as planned.

Implementation:

- Create and manage patient records.
- Verify diagnostic test records and selection.
- Verify appointment booking and management.
- Verify appointment cancellation and refund calculation.
- Verify appointment completion and report generation.
- Verify report management.
- Verify Service Portal functionality.

5.15 Conclusion

The project demonstrates a Medical Diagnosis Center application developed using ServiceNow Service Portal. The implemented solution supports patient registration, diagnostic test management, appointment booking and management, appointment cancellation, refund calculation, automatic report generation, report management, and Service Portal access. By centralizing healthcare diagnostic activities on ServiceNow, manual clerical overhead is eliminated, appointment scheduling conflicts are prevented, and the patient experience is modernized through instant self-service and automated clinical report delivery.

— End of Project Development Phase Document —
Medical Diagnosis Center (ServiceNow)